



PulseShare

Share the Pulse, Save the Life

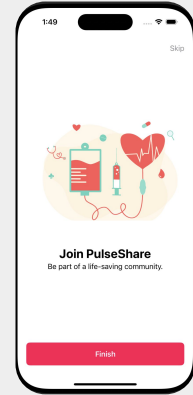
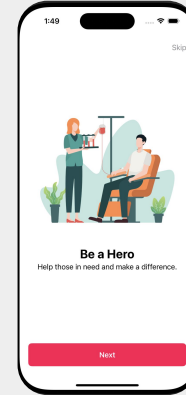
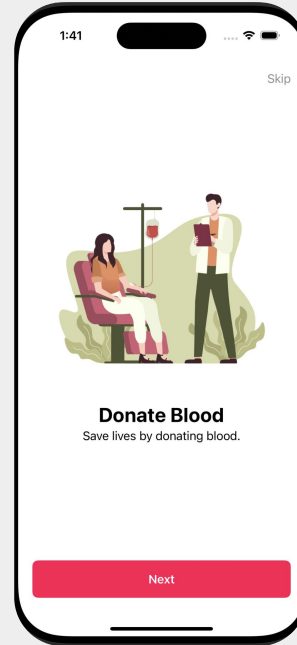
Introduction

PulseShare

Share the Pulse, Save the Life

A mobile application for blood donations.

It offers donors a platform to easily donate blood and healthcare workers to seamlessly manage the system.



Elevator Pitch

Maintaining a stable blood donor pool is a critical challenge in healthcare, with Canadian Blood Services needing 100,000 new donors annually to meet demand. Despite the altruistic motivation of most donors, organizations struggle to engage, retain, and inform donors effectively. This creates a pressing need for an accessible and impactful solution.

PulseShare addresses this challenge by simplifying the blood donation process through a mobile and web application. It streamlines appointment bookings, enhances communication with donors, and provides personalized insights into their contributions, such as the institutions receiving their donations and the lives impacted.

By fostering stronger relationships with donors and encouraging consistent participation, PulseShare aims to bridge the gap between the rising need for blood and the current donor base, creating a sustainable and scalable impact in healthcare.

Motivation

Half of all Canadians will either need blood or know someone who will need it at some point in their lives. Yet, only 4% of Canadians donate. This significant gap drives our mission to make a difference.

With PulseShare, we aim to transform the donor experience, making it simpler and more meaningful. By streamlining appointment booking, improving communication, and providing personalized insights into the impact of their contributions, we seek to inspire more Canadians to become regular donors.

Our goal is to support healthcare systems by increasing donor participation and ensuring a stable blood supply, ultimately saving lives and building a healthier community.

Purpose and Objectives

The purpose of PulseShare is to address the critical need for blood donations by simplifying the donor journey and fostering stronger connections between donors and healthcare organizations. By providing an intuitive and impactful platform, PulseShare aims to increase donor engagement, retention, and awareness, ultimately contributing to a more reliable blood supply system.

Objectives

- **Streamline Appointment Booking:** Provide an efficient and user-friendly process for scheduling and managing blood donation appointments.
- **Enhance Donor Engagement:** Strengthen relationships with donors by offering personalized insights into their contributions and the impact of their donations.
- **Promote Regular Donations:** Encourage consistent blood donations through targeted features and meaningful engagement strategies.

Problem Statement and Background

Problem Statement and Background

Maintaining a sufficient blood donor pool is a critical challenge, with only 4% of Canadians donating blood, despite half of the population needing blood or knowing someone who will. Canadian Blood Services requires 100,000 new donors annually to meet demand, yet donor engagement and retention remain issues.

Research shows that most donors are motivated by altruism, and providing insights into how their contributions are used strengthens relationships and encourages regular donations.

PulseShare addresses these challenges by simplifying appointment bookings, enhancing communication, and providing personalized insights into donation impact. This innovative solution bridges the gap between donor participation and the growing need for blood, ensuring a sustainable and reliable blood supply.

Proposed Solution

Solution Statement

PulseShare is a mobile and cloud-based application designed to improve blood donation management and donor retention. The solution combines an iOS app for donors and a Next.js-based web application for healthcare providers.

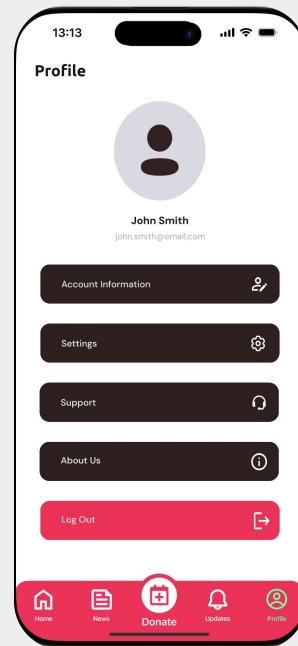
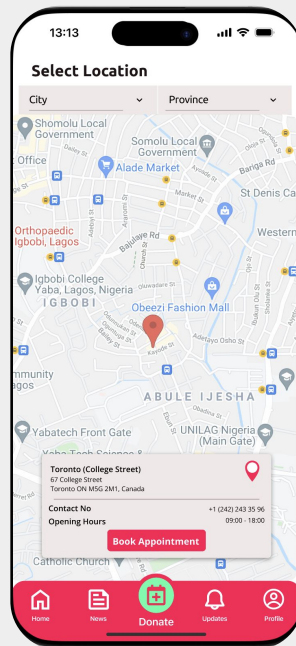
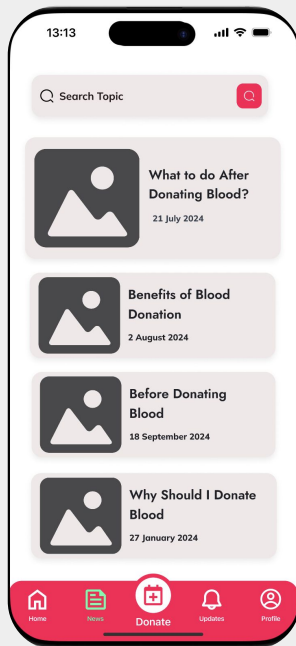
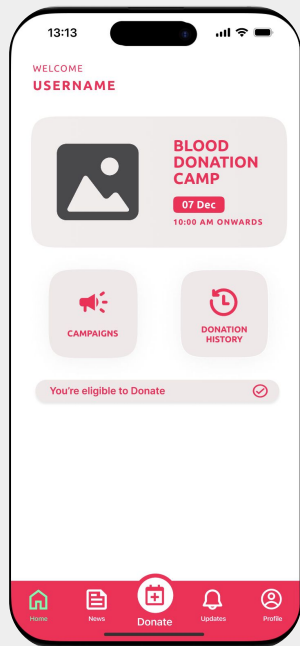
The iOS app allows donors to schedule appointments, receive reminders, track their donation history, and view personalized impact reports, offering a seamless and user-friendly experience.

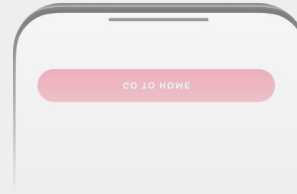
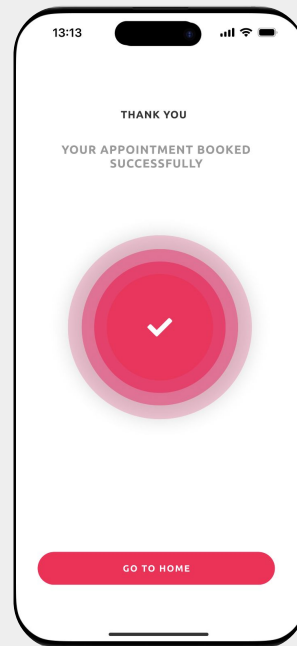
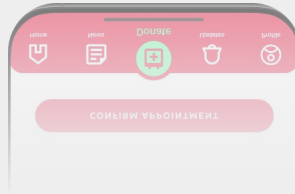
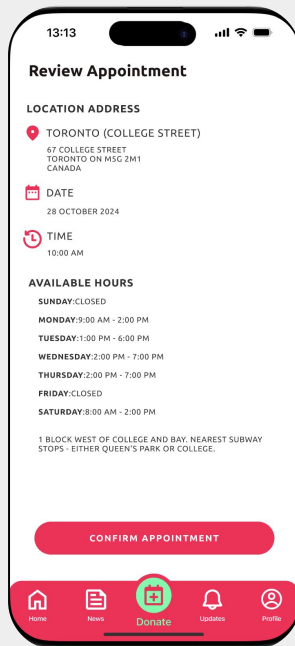
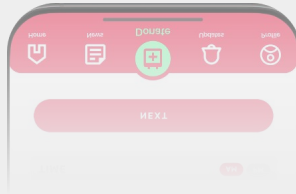
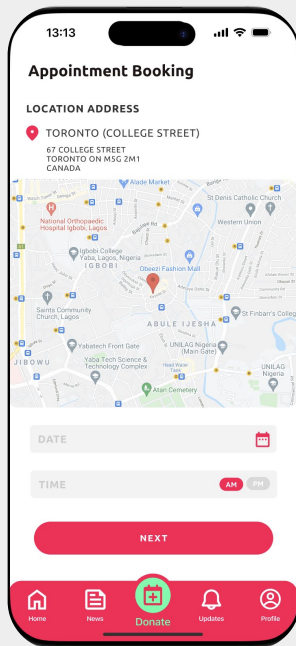
The web application provides healthcare staff with real-time access to donor data, enabling efficient management of blood supplies and enhanced coordination between donors and healthcare facilities.

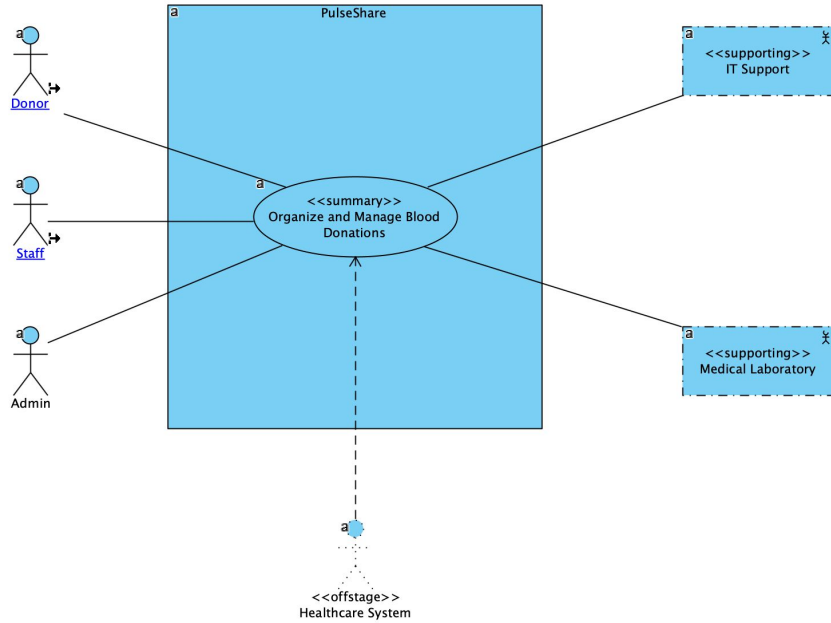
By leveraging mobile computing and cloud infrastructure, PulseShare streamlines the blood donation process and fosters stronger donor engagement.

Development Process

Wireframing and Prototyping

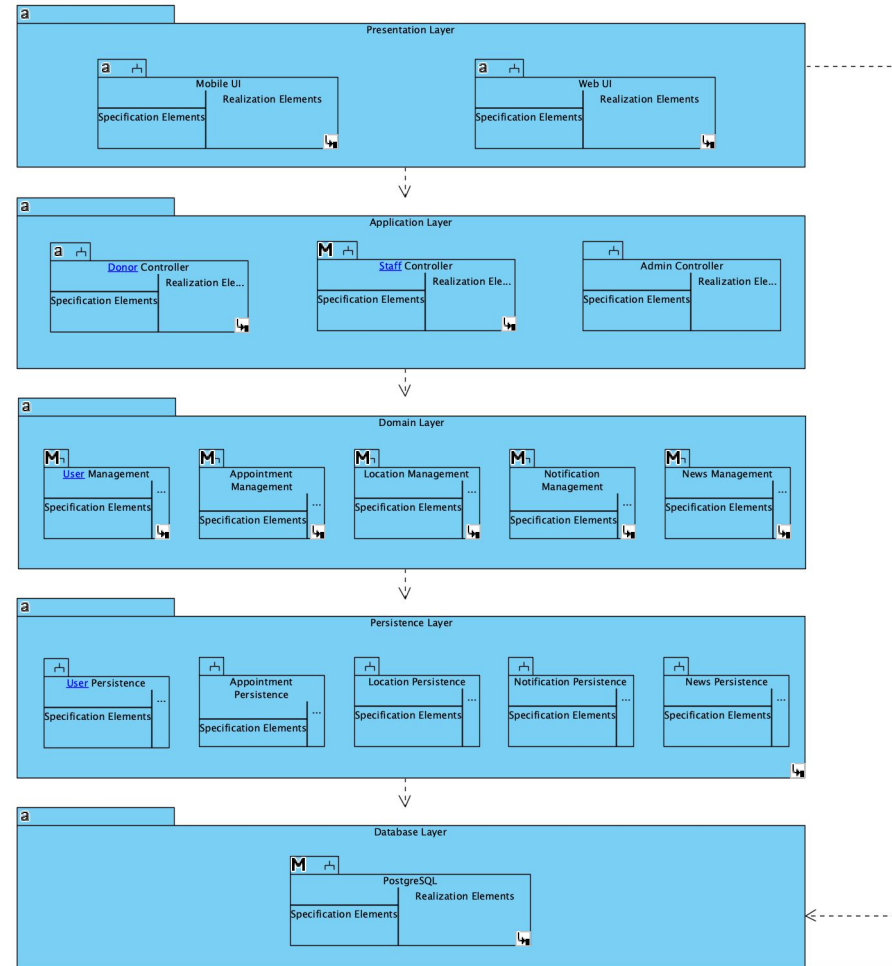


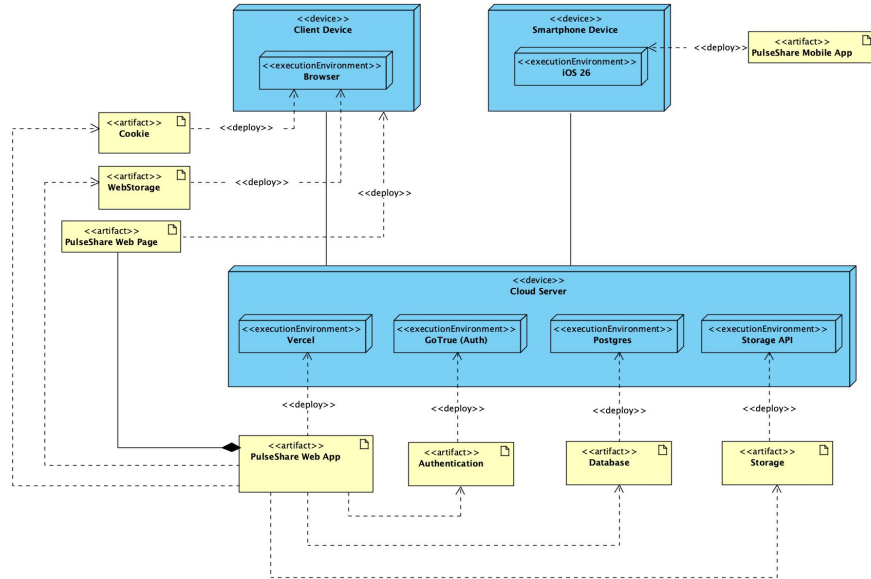




System Context

System Architecture





Deployment Model

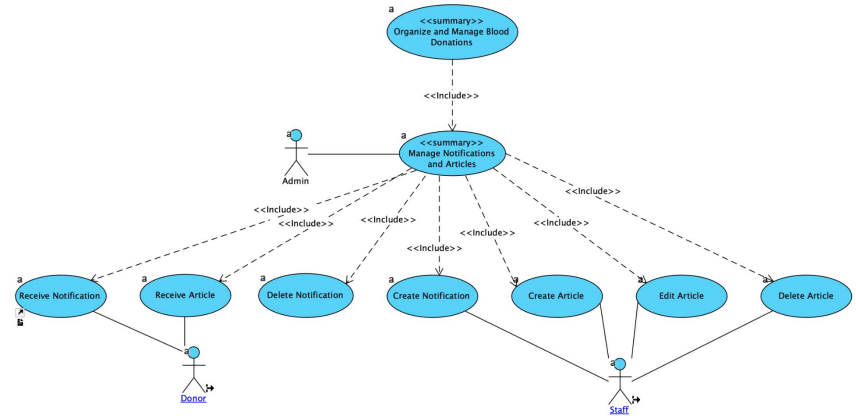
Use Case: Manage Users





Use Case: Managing Appointments and Locations

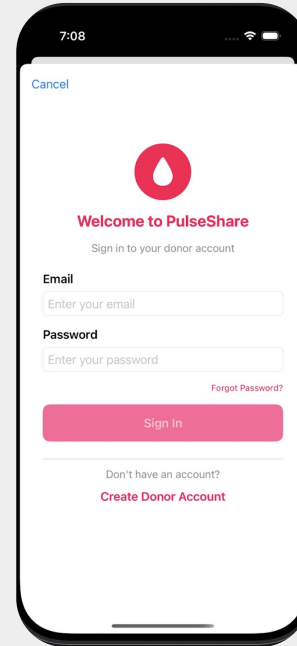
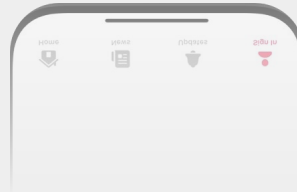
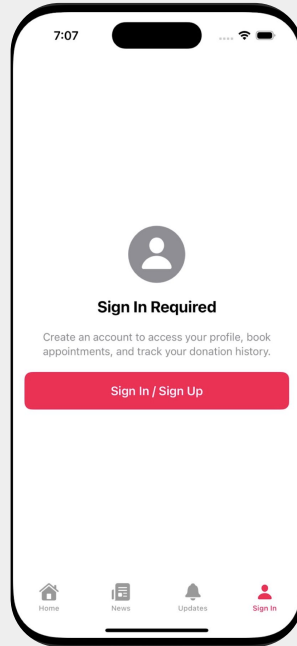
Use Case: Managing Notifications and Articles



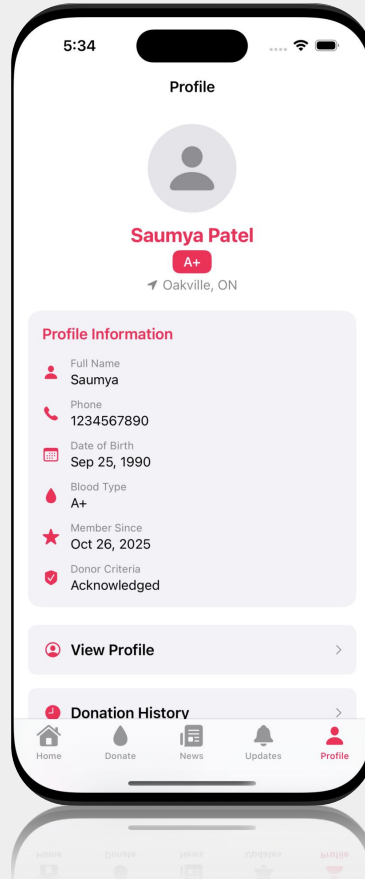
iOS Interface

User Registration & Authentication

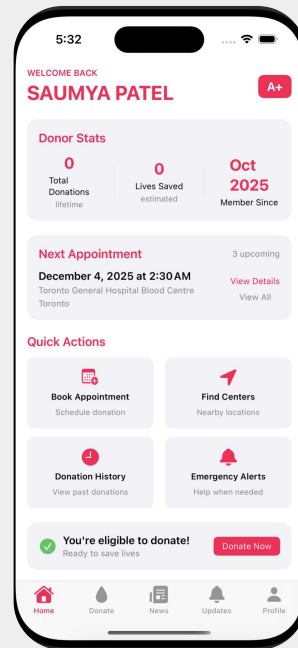
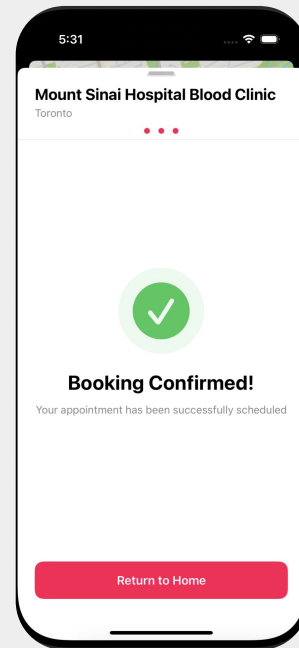
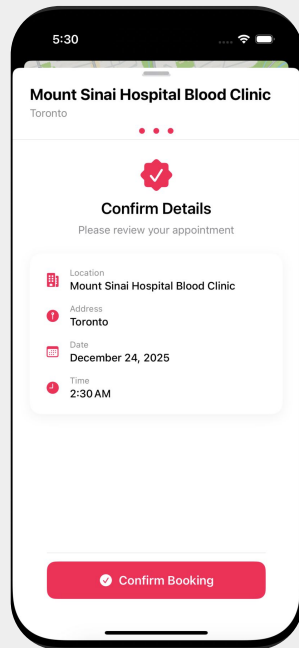
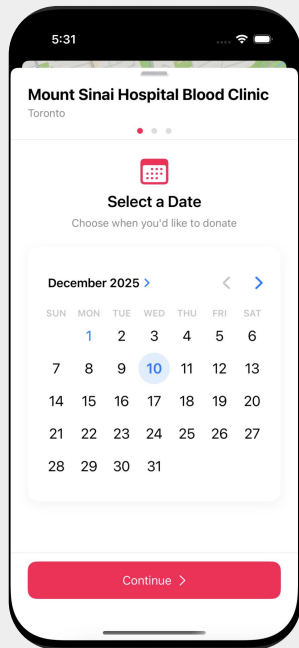
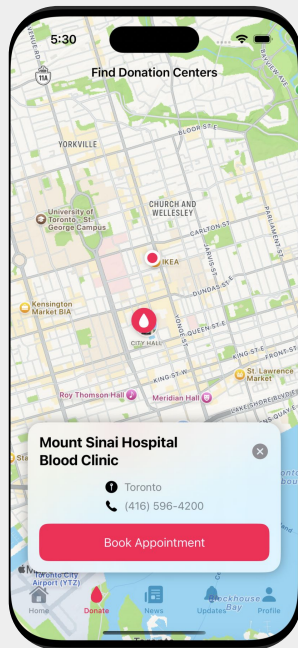
using Auth



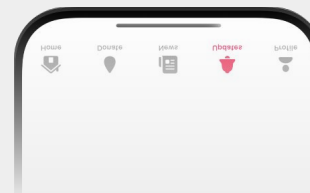
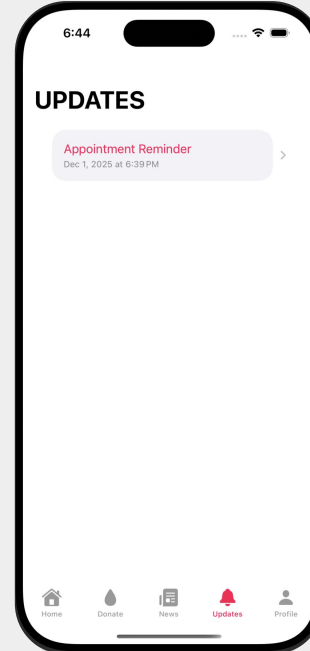
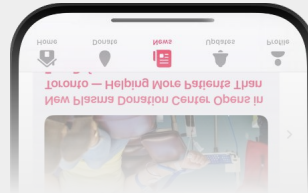
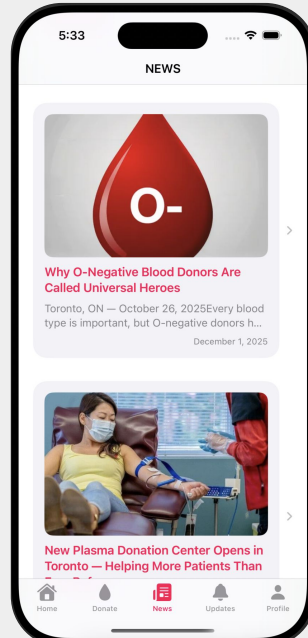
User Profile



Blood Donation Appointment Booking Procedure

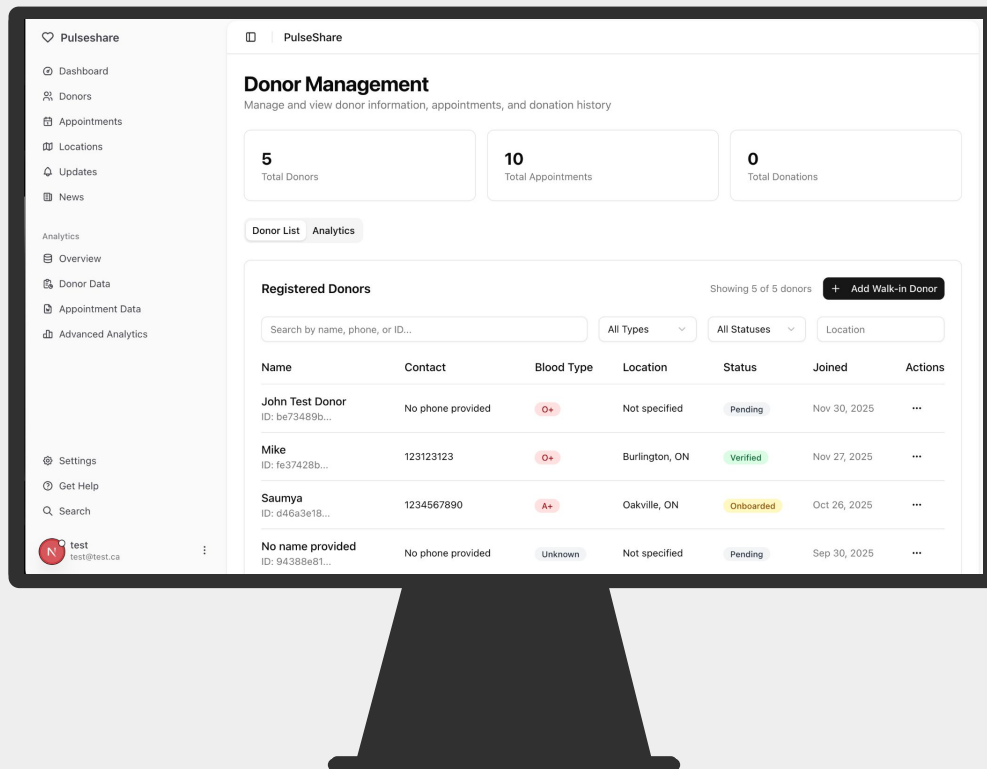


News and Updates



Web Interface

Managing Donors



Managing Appointments

PulseShare

Dashboard
Donors
Appointments
Locations
Updates
News
Analytics
Overview
Donor Data
Appointment Data
Advanced Analytics
Settings
Get Help
Search

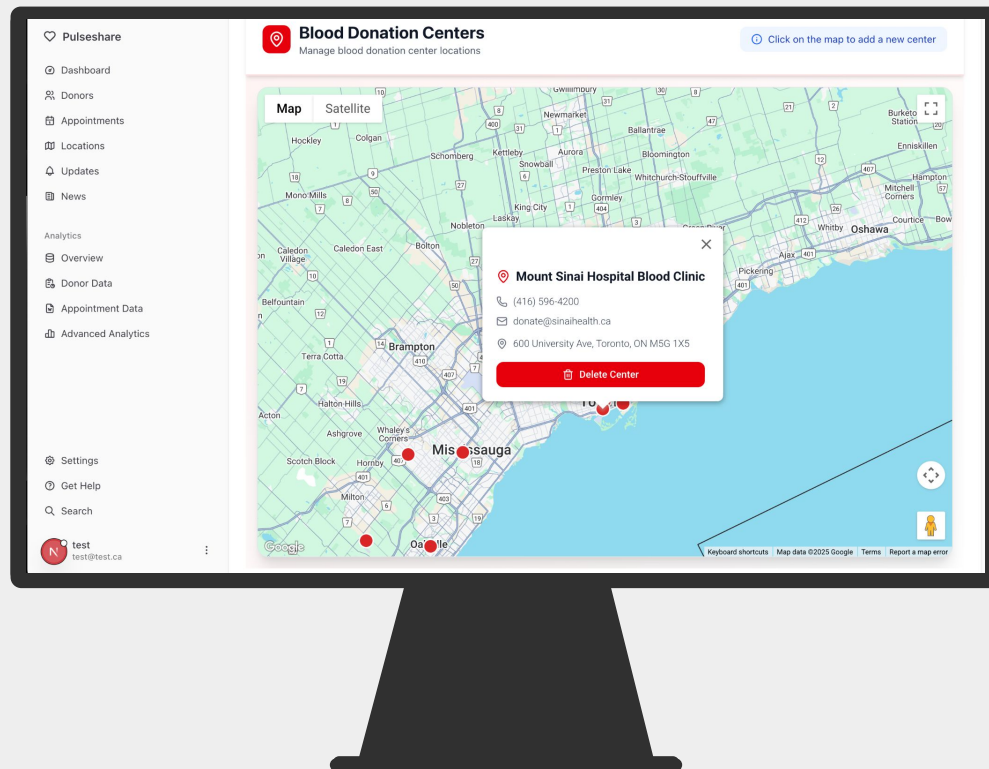
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test@test.ca

PulseShare

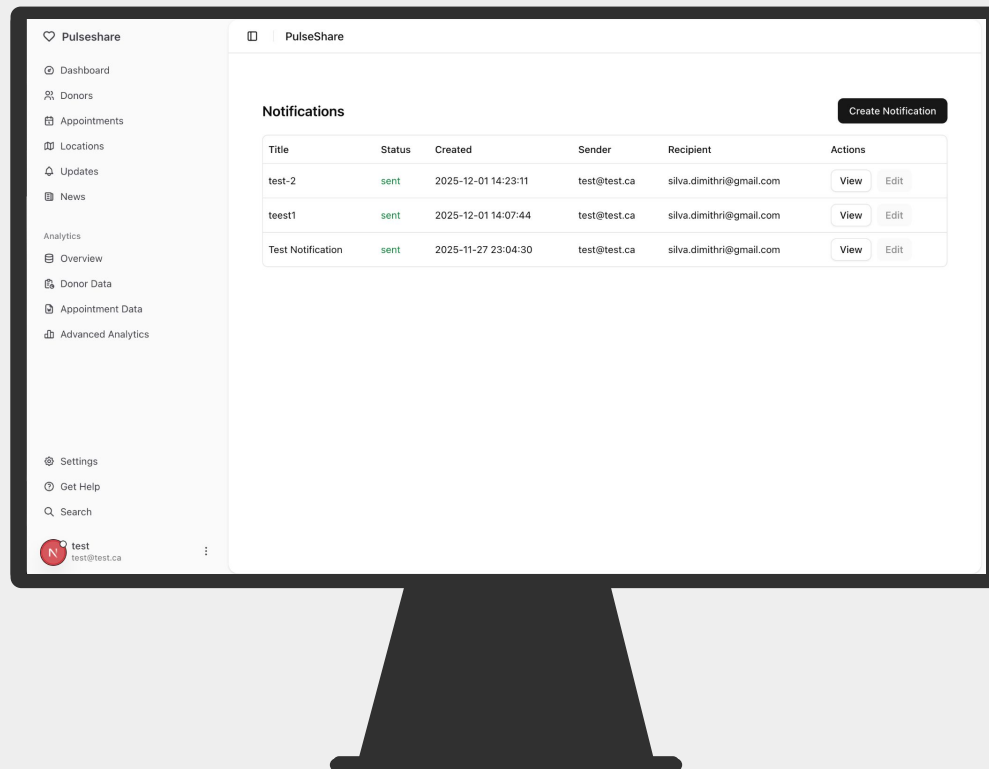
Appointments

Blood Center	Donor	Blood Type	Status	Scheduled At	Actions
Mount Sinai Hospital Blood Clinic	Saumya Patel pasaumya@sheridancollege.ca	A+	Confirmed	2025-12-10 19:00:00	View Details
Test Blood Center	John Donor donor@test.com	O+	Checked In	2025-11-30 22:52:35	View Details
Test Blood Center	John Donor donor@test.com	O+	Checked In	2025-11-30 21:41:59	View Details
Mount Sinai Hospital Blood Clinic	Saumya Patel pasaumya@sheridancollege.ca	A+	Completed	2025-12-25 23:30:00	View Details
Toronto General Hospital Blood Centre	Saumya Patel pasaumya@sheridancollege.ca	A+	Completed	2025-12-04 02:30:00	View Details
Mount Sinai Hospital Blood Clinic	Mike mikhailajaj@gmail.com	O+	Confirmed	2025-11-30 20:30:00	View Details
Princess Margaret Cancer Centre	—	—	Pending	2025-11-27 16:30:12	View Details
Mount Sinai Hospital Blood Clinic	Saumya Patel pasaumya@sheridancollege.ca	A+	Pending	2025-11-25 16:02:53	View Details
Mount Sinai Hospital Blood Clinic	Dimitri silva.dimitri@gmail.com	A+	Confirmed	2025-12-01 00:15:00	View Details
Princess Margaret Cancer Centre	Saumya Patel pasaumya@sheridancollege.ca	A+	Pending	2025-11-22 16:11:51	View Details

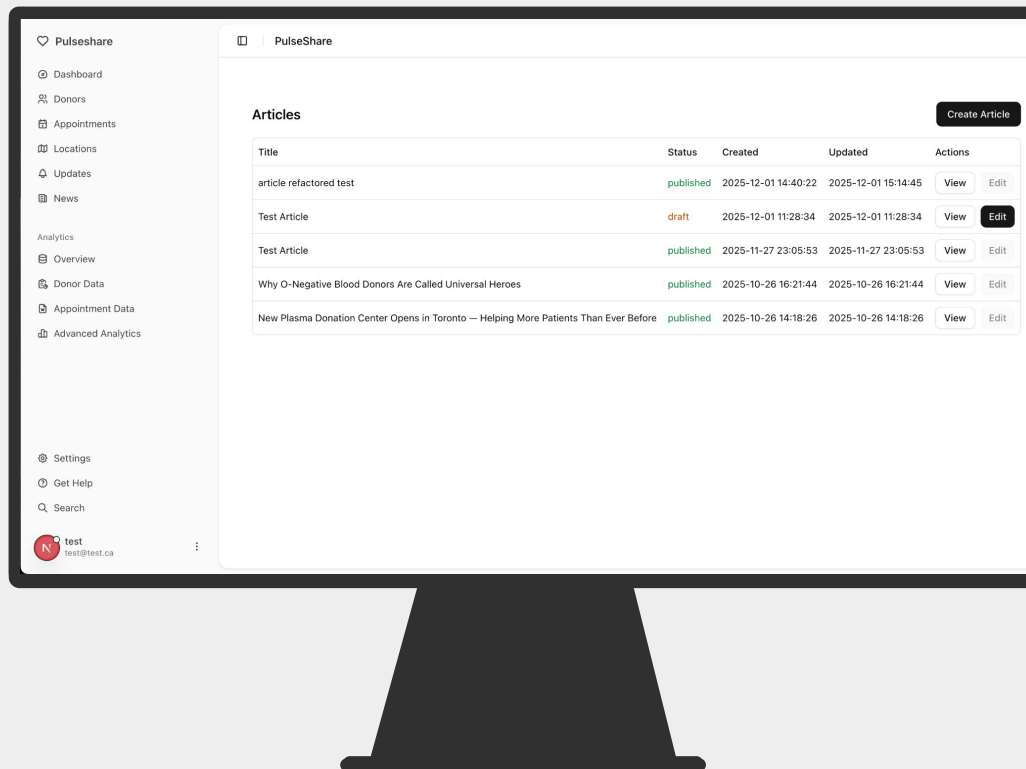
Managing Locations



Managing Updates

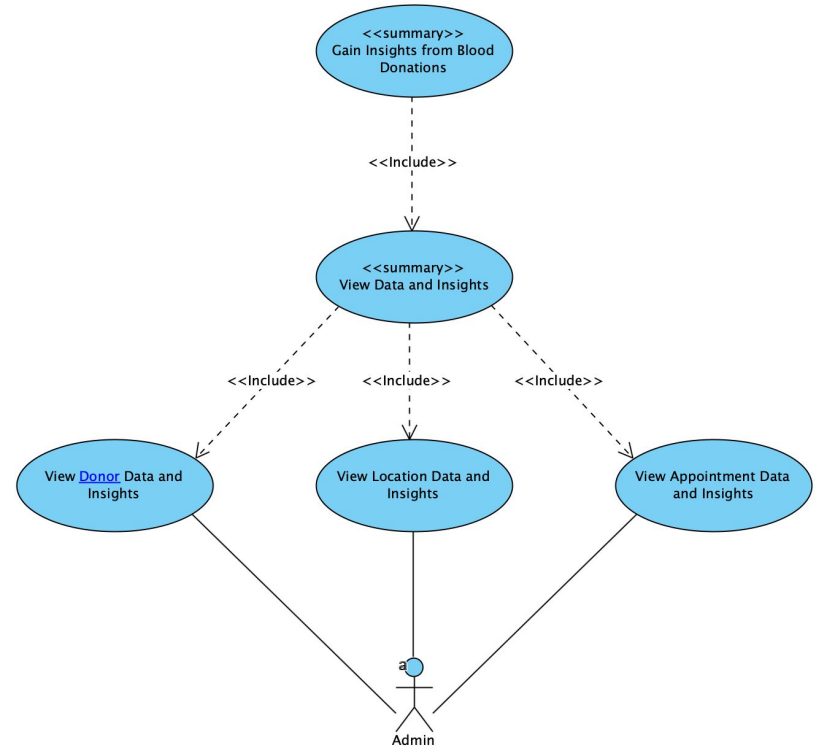


Managing News Articles

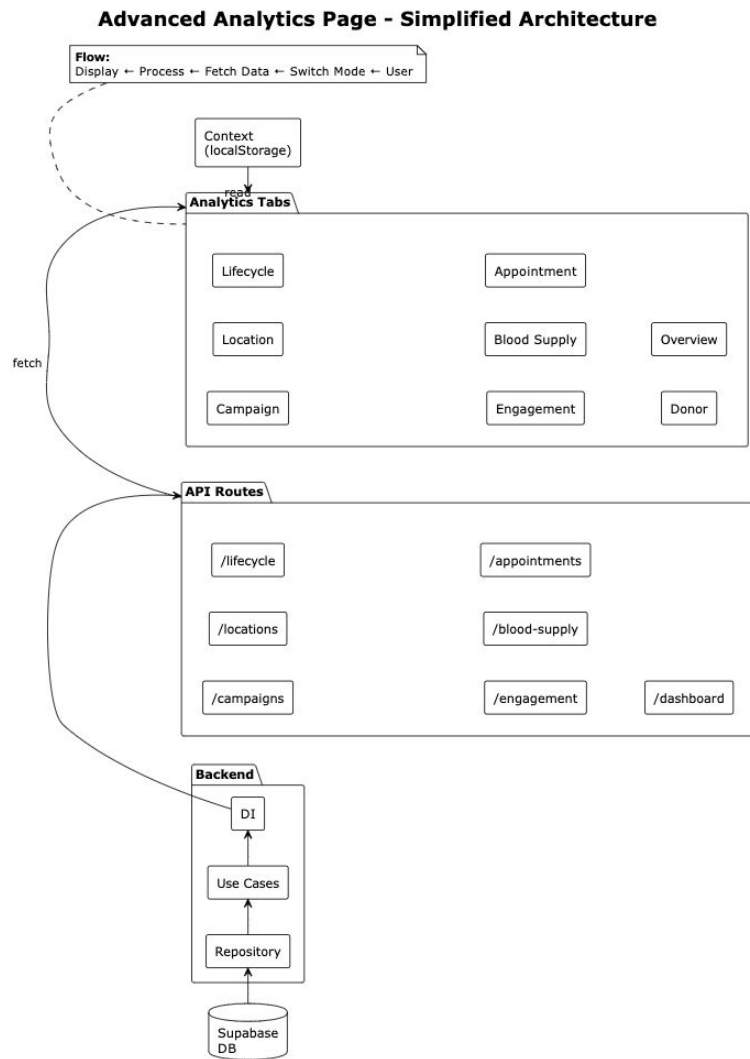


Advanced Computer Science Areas

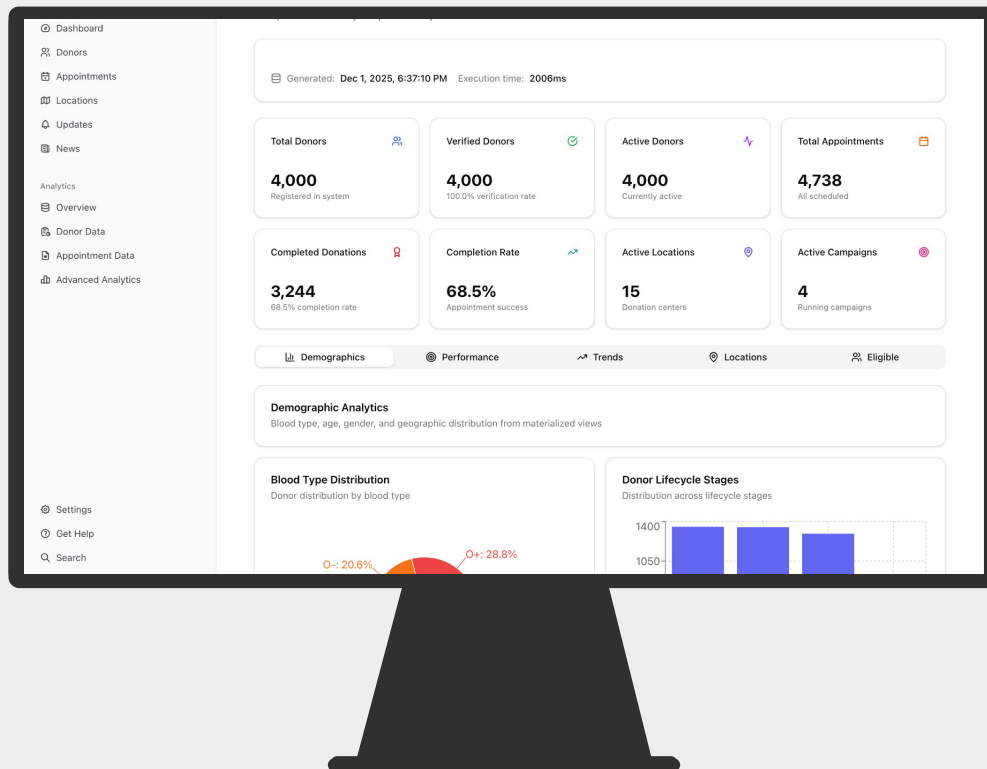
Data Analytics: Gain Insights from Blood Donations

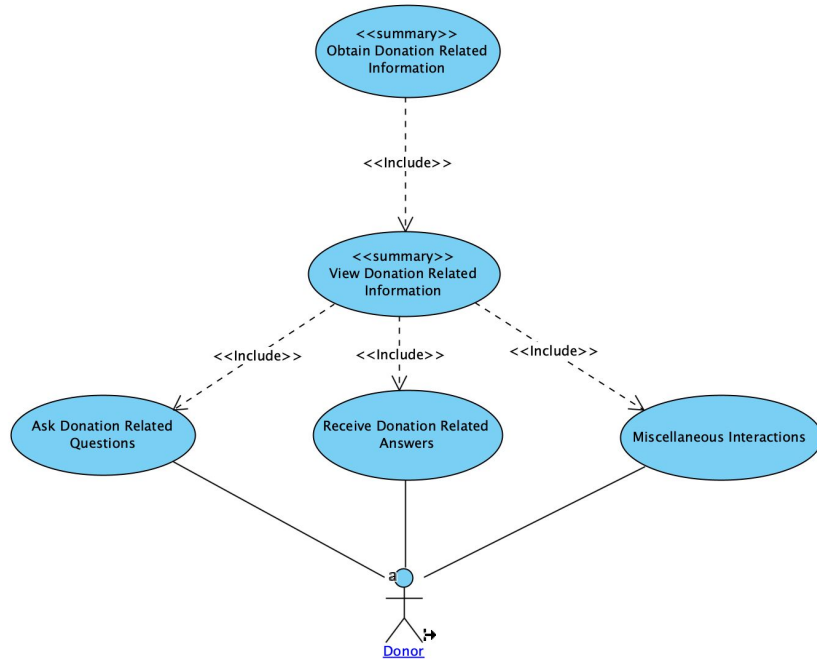


Data Analytics: Gain Insights from Blood Donations using Amazon Redshift



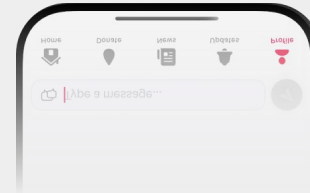
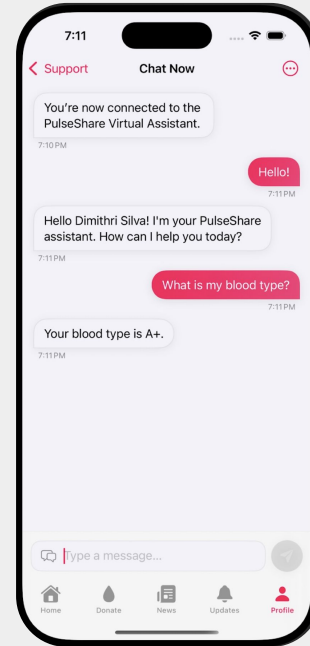
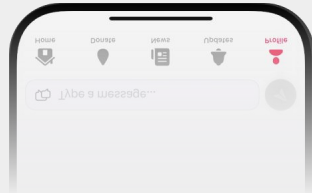
Data Analytics View





**Cognitive Computing:
Receive Support from
the PulseShare
Virtual Assistant**

PulseShare Virtual Assistant



Thank You